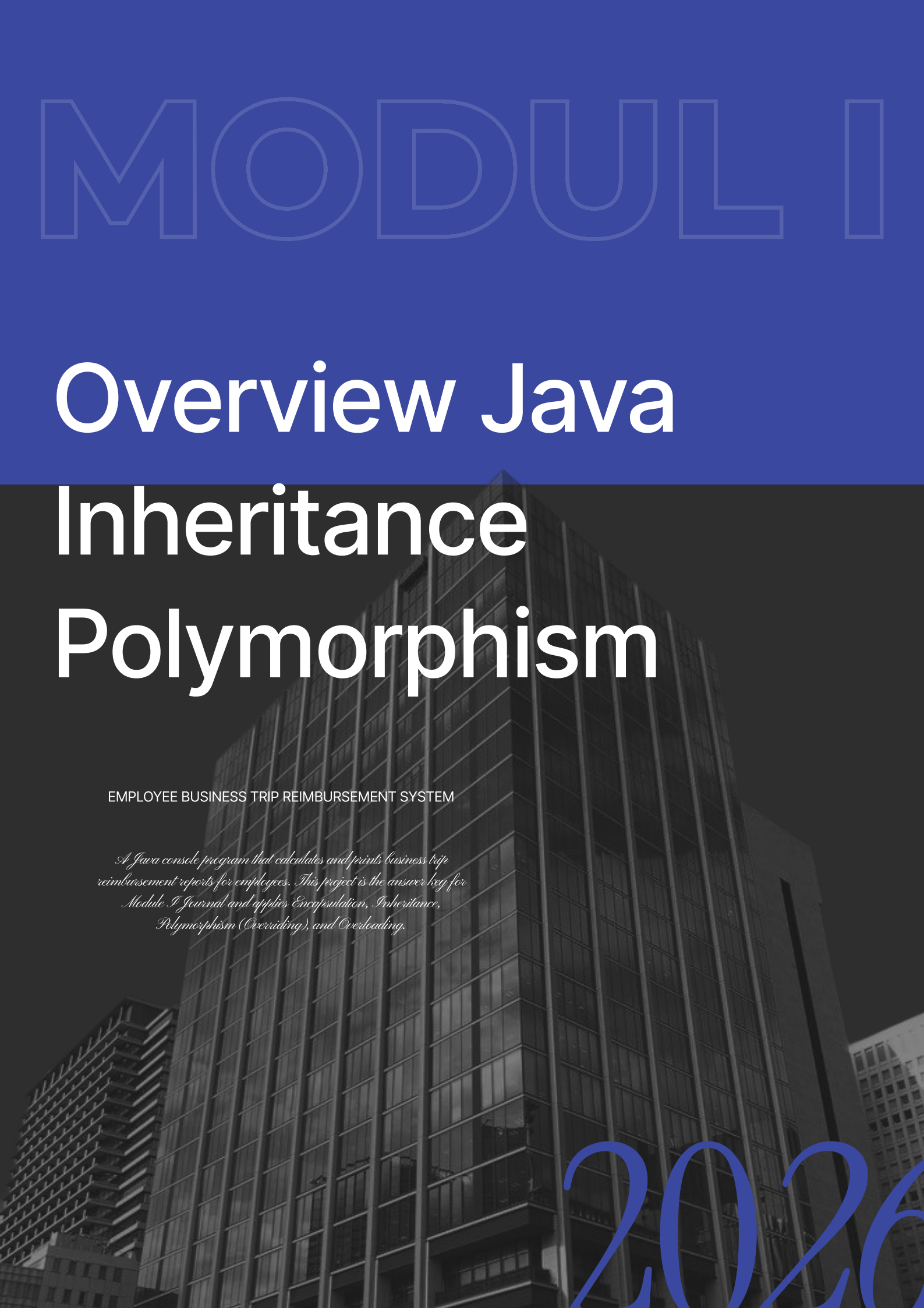
MODUL I

Overview Java Inheritance Polymorphism

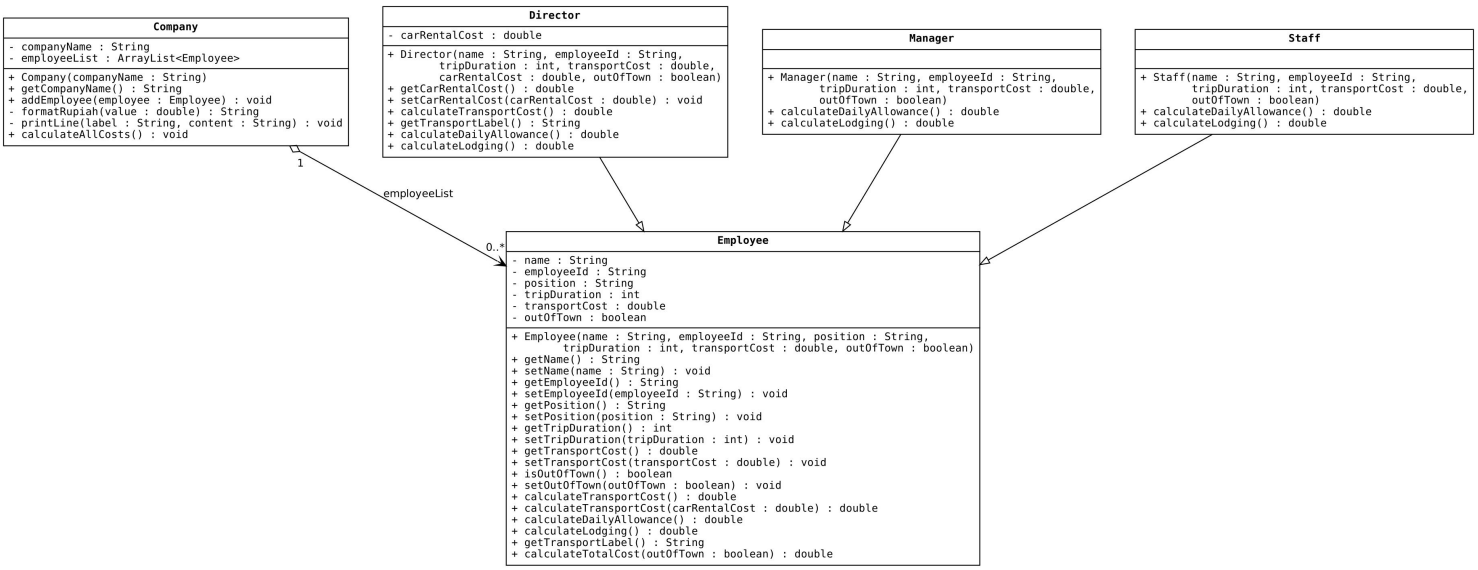
EMPLOYEE BUSINESS TRIP REIMBURSEMENT SYSTEM

A Java console program that calculates and prints business trip reimbursement reports for employees. This project is the answer key for Module I Journal and applies Encapsulation, Inheritance, Polymorphism (Overriding), and Overloading.



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Study Case



A consulting company named "PT Mitra Konsultan Prima" often sends its employees on business trips. There are 3 employee positions that are entitled to submit a reimbursement: Director, Manager, and Staff.

Every business trip is paid for by the company and consists of a Daily Allowance, Lodging Cost, and Transport Cost. The daily allowance and the lodging rate differ for each position. Lodging cost is covered only if the trip destination is Out of Town.

If the destination is In Town, the employee does not stay overnight, so no lodging cost is paid. Specifically for the Director, the company also covers a Car Rental Cost, which is added to the Transport Cost.

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Rules

Daily Allowance (per trip day):

- Director : Rp 500.000
- Manager : Rp 350.000
- Staff : Rp 200.000

Lodging Cost (per night, number of nights = trip duration - 1) :

- Director : Rp 1.200.000
- Manager : Rp 800.000
- Staff : Rp 400.000

Transport Cost:

- Director : ticket cost + car rental cost
- Manager : ticket cost
- Staff : ticket cost



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Required Methods:

calculateTransportCost()	Returns the standard transport cost (ticket cost).
calculateTransportCost(double carRentalCost))	Overloading. Adds the car rental cost to the standard transport cost (used only by the Director).
calculateDailyAllowance()	Calculates the allowance for the whole trip based on the position (overriding).
calculateLodging()	Calculates the lodging cost based on the position and number of nights (overriding).
CalculateTotalCost(boolean outOfTown)	Calculates the total reimbursement: Daily Allowance + Transport Cost + Lodging Cost (only if the destination is Out of Town).
calculateAllCost()	A method in the Company class that loops through and prints the reimbursement report of all employees stored in the ArrayList, then prints the grand total of all reimbursements.

Employee List

01 Director named Bayu (ID: DIR01), on a 4-day trip Out of Town. Transport cost (ticket) is Rp 3.000.000 and car rental cost is Rp 1.500.000.

02 Manager named Maya (ID: MGR01), on a 3-day trip Out of Town. Transport cost (ticket) is Rp 1.800.000.

03 Staff named Fajar (ID: STF01), on a 5-day trip Out of Town. Transport cost (ticket) is Rp 1.200.000.

04 Staff named Nadia (ID: STF02), on a 2-day trip In Town. Transport cost is Rp 150.000.



Task

Create a Java program that implements Encapsulation, Inheritance, Polymorphism (Overriding), and Overloading according to the following requirements:



1. Create the *Employee* class as the superclass. All of its attributes are private and are accessed through getter and setters. This class is not an abstract class.
2. Create classes for *Director, Manager, Staff* as subclass of *Employee*. Subclass-specific attributes (the Director's car rental cost) are also private and have their own getter and setter.
3. Create the *Company* class, which stores all employees in an ArrayList.
4. Create the *Main* class containing the main method that creates the employee data and prints the reimbursement report.

Main.*java*

```java
public class Main {
    public static void main(String[] args) {
        Company company = new Company("PT Mitra Konsultan Prima");

        Employee employee1 = new Director("Bayu", "DIR01", 4, 3000000, 1500000, true);
        Employee employee2 = new Manager("Maya", "MGR01", 3, 1800000, true);
        Employee employee3 = new Staff("Fajar", "STF01", 5, 1200000, true);
        Employee employee4 = new Staff("Nadia", "STF02", 2, 150000, false);

        company.addEmployee(employee1);
        company.addEmployee(employee2);
        company.addEmployee(employee3);
        company.addEmployee(employee4);

        company.calculateAllCosts();
    }
}
```

Output *Console*:

```
=====================================
BUSINESS TRIP REIMBURSEMENT REPORT
=====================================
Employee Name    : Bayu
ID               : DIR01
Position         : Director
Trip Duration    : 4 days
Destination      : Out of Town
Daily Allowance  : Rp 2.000.000
Lodging          : Rp 3.600.000
Transport + Car Rental : Rp 4.500.000
Total Reimbursement : Rp 10.100.000
-------------------------------------
Employee Name    : Maya
ID               : MGR01
Position         : Manager
Trip Duration    : 3 days
Destination      : Out of Town
Daily Allowance  : Rp 1.050.000
Lodging          : Rp 1.600.000
Transport Cost   : Rp 1.800.000
Total Reimbursement : Rp 4.450.000
-------------------------------------
Employee Name    : Fajar
ID               : STF01
Position         : Staff
Trip Duration    : 5 days
Destination      : Out of Town
Daily Allowance  : Rp 1.000.000
Lodging          : Rp 1.600.000
Transport Cost   : Rp 1.200.000
Total Reimbursement : Rp 3.800.000
-------------------------------------
Employee Name    : Nadia
ID               : STF02
Position         : Staff
Trip Duration    : 2 days
Destination      : In Town
Daily Allowance  : Rp 400.000
Lodging          : Rp 0 (no overnight stay)
Transport Cost   : Rp 150.000
Total Reimbursement : Rp 550.000
=====================================
GRAND TOTAL      : Rp 18.900.000
=====================================
```

Method for rupiah *format*:

```java
// Format rupiah value: 6000000 -> "Rp 6.000.000"
private String formatRupiah(double value) {
    long rounded = Math.round(value);
    String digits = String.valueOf(Math.abs(rounded));
    StringBuilder result = new StringBuilder();
    for (int i = 0; i < digits.length(); i++) {
        if (i > 0 && (digits.length() - i) % 3 == 0) {
            result.append('.');
        }
        result.append(digits.charAt(i));
    }
    return (rounded < 0 ? "-Rp " : "Rp ") + result;
}
```